MISTRO: A Trust-Aware Digital Marketplace for Reliable Home-Service Professional Matching

Humanized IEEE-Style Research Paper Draft

Project: MISTRO — “Service On, Stress Gone”
Basis: Prepared from the user-provided MISTRO project presentation (Sairam SDG Solveathon 4.0).

Drafting note. This manuscript has been written from scratch in original academic wording based on the supplied project material. It does not intentionally reproduce sentences from the presentation. A plagiarism or similarity percentage cannot be guaranteed because technical terminology, project names, and standard academic expressions may naturally overlap with other documents. Before submission, external claims and references should be independently verified and the final manuscript should be checked with the target venue's similarity policy.

Abstract

Finding a dependable professional for an urgent household repair is often more difficult than the repair itself. Customers may rely on personal recommendations, general web searches, or informal local contacts, while skilled professionals may depend on word-of-mouth networks and face inconsistent access to new jobs. MISTRO proposes a two-sided digital marketplace that brings these two groups together through a structured service discovery, verification, matching, and booking process. The platform is designed to help customers identify professionals according to service skill, location, availability, and reputation, while giving professionals a dedicated environment for handling jobs, earnings, and their digital reputation. Trust is treated as a core part of the system through identity and professional verification mechanisms, including Aadhaar-based identity checks, trade-license checks, and photo identification as described in the project design. The project also explores an AI-assisted layer organized around four areas—service, price, trust, and verification—to support decision-making throughout the service journey. A prototype was developed to demonstrate the proposed marketplace and its user workflows. The work presented here focuses on the system concept, architecture, matching approach, trust model, prototype, preliminary user validation, and directions for future quantitative evaluation.

Keywords— Home-service marketplace, trust-aware matching, professional verification, service recommendation, artificial intelligence, digital marketplace, reputation system.

I. INTRODUCTION

Household services such as electrical repair, plumbing, carpentry, painting, cleaning, and appliance maintenance are practical needs that arise regularly. Despite this recurring demand, finding the right professional can still involve a surprisingly informal process. A customer may call several people, ask friends for recommendations, search online, compare uncertain information, and wait for callbacks before a suitable person is found. The situation becomes more difficult when the requirement is urgent or when the customer has little prior knowledge of the service provider.

The problem is not limited to customers. Skilled professionals can also struggle to maintain a steady flow of work when their business depends mainly on local contacts and referrals. A capable electrician or plumber may have the technical ability to solve a problem but still have limited visibility beyond an existing network. The absence of a structured digital reputation and job-management environment can further restrict opportunities.

MISTRO was conceived to address this two-sided problem through a mobile-first marketplace. The project presentation describes the platform as a system that connects homeowners and renters with verified home-service professionals and supports browsing, comparison, booking, communication, scheduling, and professional management. The intended matching process uses service skill, location, availability, and ratings as key inputs. filecite turn3file3

Rather than treating trust as an optional feature added after a booking is made, MISTRO places verification and reputation near the center of the service-selection process. The project also introduces an AI-oriented

layer covering service, price, trust, and verification. These areas provide a direction for intelligent assistance, while the present paper carefully distinguishes the demonstrated prototype from functionality that remains to be implemented or experimentally validated.

A. Contributions

- A two-sided marketplace model that serves both customers and service professionals.
- A multi-factor matching concept using skill, location, availability, and customer ratings.
- A verification-oriented trust layer based on identity and professional credentials.
- Separate customer and professional workflows through the MISTRO and MistroCrew interfaces.
- An AI-assisted framework organized around service, price, trust, and verification.
- A prototype and preliminary user-validation material that provide a basis for further technical evaluation.

II. PROBLEM STATEMENT AND MOTIVATION

The project identifies a gap between the demand for household services and the way customers and professionals currently find one another. Customers may face uncertainty about whom to contact, whether the person is genuine, whether the required skill is available, and whether the quoted price is reasonable. Professionals, meanwhile, may experience irregular bookings and limited reach.

A. Customer-Side Challenges

- Time spent searching and contacting multiple professionals.
- Limited visibility into professional identity and qualifications.
- Difficulty comparing providers using consistent information.
- Concerns about overcharging or unclear pricing.
- Limited accountability after a service is completed.
- Difficulty obtaining an appropriate professional for urgent requirements.

B. Professional-Side Challenges

- Irregular access to service requests.
- Dependence on word-of-mouth and local networks.
- Limited mechanisms for demonstrating a reliable service history.
- Fragmented management of bookings, jobs, and earnings.
- Difficulty reaching customers outside an existing network.

The MISTRO presentation summarizes the opportunity as the need for a centralized, verified, trust-oriented platform that can connect skilled professionals with homeowners in Tier 1 and Tier 2 cities. □filecite□turn3file7□

III. EXISTING APPROACHES

The project presentation identifies several ways in which customers currently search for home-service providers. These include word-of-mouth recommendations, general search engines, established home-service platforms, and local service listings. Each approach can be useful, but the information and level of trust available to the customer can vary substantially.

Word-of-mouth recommendations can provide a degree of social trust, but they are naturally limited by the size of a person's network. General online searches increase reach but can require substantial manual comparison and may not provide a standardized view of professional credentials. The project also compares MISTRO with established marketplace alternatives and identifies differences in transaction handling, verification, positioning, and the role of independent professionals. These comparisons should be supported with independently verified sources in the final publication.

The proposed contribution of MISTRO is therefore not simply another directory of service providers. Its central idea is to combine discovery, verification, matching, booking, communication, and reputation within one coordinated workflow.

IV. PROPOSED MISTRO SYSTEM

MISTRO is organized as a two-sided service marketplace. On one side, customers describe or select the service they require and search for suitable professionals. On the other side, professionals receive opportunities and manage the jobs associated with their profiles. The project uses two complementary interfaces: MISTRO for customers and MistroCrew for professionals. □filecite□turn3file7□

A. Customer Application

- Browse and compare home-service categories and professionals.
- View relevant professional information before booking.
- Select available time slots and create service bookings.
- Communicate with the assigned professional through the platform.
- Provide ratings and reviews after service completion.

B. Professional Application

- Receive and manage customer service requests.
- Organize upcoming and completed jobs.
- Track earnings and service activity.
- Maintain a professional profile and build a reputation.

C. End-to-End Workflow

A typical service journey can be represented as: Requirement → Service Identification → Candidate Discovery → Trust/Verification Review → Matching → Booking → Service Execution → Feedback. This sequence is intended to reduce the amount of manual effort required from the customer while creating a repeatable process for the professional.

V. SYSTEM ARCHITECTURE

At a conceptual level, the system can be divided into five cooperating layers: the customer interface, the professional interface, the service and matching layer, the trust and verification layer, and the AI-assisted layer. The prototype screens included in the project presentation demonstrate the intended interaction flow, while the final research paper should add the actual software architecture and technology stack used in implementation. □filecite□turn8file9□

A. Service Layer

The service layer organizes household requirements into recognizable categories such as electrical work, plumbing, carpentry, painting, cleaning, and repair activities. This categorization gives the matching system a structured representation of what the customer needs.

B. Matching Layer

The matching layer receives the service requirement and identifies professionals whose profiles satisfy the relevant constraints. The project specifies four principal matching dimensions: skill, location, availability, and ratings. □filecite□turn3file7□

C. Trust Layer

The trust layer is responsible for presenting evidence that helps a customer evaluate a professional. The project design includes Aadhaar-based identity verification, trade-license checks, photo identification, and reviews. □filecite□turn3file0□

D. Communication and Booking Layer

Once a suitable professional is selected, the platform is designed to support scheduling and direct communication before and during the service. This reduces the need to move the interaction to disconnected communication channels.

E. AI-Assisted Layer

The presentation identifies four AI-oriented areas—SERVICE, PRICE, TRUST, and VERIFY. □filecite□turn6file0□ In the current manuscript, these are treated as an AI-assisted design direction unless the underlying implementation is demonstrated and measured. This distinction prevents conceptual functionality from being presented as a validated machine-learning result.

VI. TRUST-AWARE MULTI-FACTOR MATCHING

A central idea of MISTRO is that a suitable professional should not be selected using service category alone. A customer who needs a plumber at a particular time may benefit more from a nearby available plumber with a strong service history than from an arbitrary plumber who happens to belong to the same category.

Let a customer request be represented as R and the set of available professionals as P = {p1, p2, ..., pn}. For each professional p, four compatibility dimensions can be considered: skill compatibility, location suitability, availability, and reputation.

A general research formulation for a ranking score is shown below. This equation is a proposed formulation unless the exact weighting and scoring method is implemented in the final system.

$$S(p,R) = w_s S_s + w_l S_l + w_a S_a + w_r S_r$$

Here, S_s represents skill compatibility, S_l represents location suitability, S_a represents availability, and S_r represents reputation. The weights determine how strongly each factor contributes to the ranking. A future experimental study can compare fixed weighting, learned ranking, and machine-learning-based recommendation approaches using real or appropriately anonymized booking data.

VII. PROFESSIONAL VERIFICATION AND TRUST

Trust is especially important when a marketplace connects customers with professionals who may be unknown to them. MISTRO therefore treats professional verification as part of the onboarding and discovery process.

The project specifies government identity verification, trade-license checks, and photo identification as verification mechanisms. It also includes ratings and dual reviews as part of the trust and transparency experience. □filecite□turn8file6□

From a research perspective, verification should be viewed as a layered process rather than as a single binary label. Identity verification answers whether a person is genuine; credential verification provides evidence of professional capability; reputation information provides a record of previous customer experience. Combining these signals can give users a richer basis for decision-making.

Any production implementation involving Aadhaar or other identity documents must additionally address consent, data minimization, secure storage, access control, retention, and applicable legal and regulatory requirements. These aspects should be documented separately in the final implementation report.

VIII. AI-ASSISTED FUNCTIONALITY

The project presentation introduces four AI-oriented agents or functional areas: Service, Price, Trust, and Verify. □filecite□turn6file0□ These can be interpreted as four points at which intelligent assistance could reduce uncertainty in the marketplace.

A. Service Assistance

A service-oriented assistant can interpret a customer's natural-language description and map it to an appropriate service category. For example, a description of a leaking pipe could be converted into a plumbing-related service request. This reduces the need for the customer to understand technical service terminology.

B. Price Assistance

A price-oriented component can eventually estimate an expected price range using factors such as service category, task complexity, location, and historical transactions. A production model would need sufficient data and careful evaluation to avoid presenting an unreliable estimate as a guaranteed price.

C. Trust Assistance

A trust-oriented component can summarize relevant verification and reputation information so that users do not need to interpret multiple indicators independently.

D. Verification Assistance

A verification-oriented component can support onboarding by organizing credential information and identifying records that require review. Automated decisions should remain subject to appropriate

safeguards and human oversight where necessary.

These AI functions represent a promising extension of the marketplace, but their research value will depend on measurable implementation. The final paper should report model type, training data, evaluation protocol, metrics, and error analysis if machine-learning models are actually deployed.

IX. PRELIMINARY USER VALIDATION

The MISTRO presentation includes a primary-survey section intended to understand whether the identified customer problem is meaningful. The presentation reports an 80% figure under the heading of market readiness and notes that respondents who managed services independently still reported previous experiences with unreliable service. filecite turn3file2

This finding is useful as an early product-validation signal, but it should not be treated as a population-level statistical result without the survey methodology. For an IEEE submission, the final paper should report the sample size, respondent profile, questionnaire, sampling procedure, collection period, and the precise definition used for the reported percentage.

The distinction matters because a product survey and a controlled research experiment answer different questions. The survey can establish whether a pain point appears meaningful among the respondents, while a technical experiment is needed to demonstrate whether the proposed matching or AI mechanism actually improves measurable outcomes.

X. MARKET CONTEXT AND BUSINESS MODEL

The project presentation estimates a Tamil Nadu market opportunity using a top-down TAM/SAM/SOM calculation. It uses a 2025 population estimate, assumptions about households requiring home services, urban population share, active use of professional services, and an initial target-city capture assumption. The resulting figures in the presentation are 3.7 crore households for TAM, 75 lakh households for SAM, and approximately 18–19 lakh households for SOM. filecite turn3file6

These values should be treated as project assumptions and estimates until the underlying demographic and market sources are cited and the calculation method is independently verified.

A. Proposed Revenue Streams

Revenue stream	Project proposal
Transaction commission	15–20%
Professional subscription	₹199 per month
Featured listing	₹499 per month
Advertising	In-app advertising

The proposed business model is a project-level plan rather than measured financial performance. The final paper should clearly distinguish projected revenue assumptions from observed results. filecite turn3file7

XI. PROTOTYPE AND IMPLEMENTATION

The project presentation contains a prototype overview and dedicated screens for dashboards and bookings. It also presents an AI-agent section associated with the Service, Price, Trust, and Verify functions. filecite turn8file9

The prototype demonstrates the intended user journey from service discovery toward booking and management. For the final IEEE manuscript, this section should be expanded with the actual implementation stack, database design, API structure, authentication method, deployment environment, and security controls used by the team. Those details should come from the source code and deployed system rather than being inferred from the presentation.

Suggested Implementation Evaluation

- Average time from service request to professional selection.
- Percentage of requests receiving at least one suitable match.

- Ranking quality measured through relevance or top-k accuracy.
- Booking completion rate.
- User-perceived usability using a standard questionnaire.
- Application response time and API latency.
- Verification success and manual-review rates.
- Cancellation and repeat-booking rates.

XII. DEPLOYMENT AND GROWTH STRATEGY

The project proposes a staged growth path consisting of Launch, Optimise, Scale, Validate, and Expand. The initial stage focuses on launching an MVP in one or two cities, collecting user feedback, and acquiring early users. Later stages focus on retention, referrals, matching improvements, trust mechanisms, expansion across major Tamil Nadu cities, and eventually entry into additional regions and service categories. filecite turn6file3

The presentation also proposes a first-year goal of 20,000 app downloads across Tamil Nadu, supported by social-media campaigns, search advertising, influencer collaborations, WhatsApp communities, local outreach, apartment demonstrations, and partnerships with hardware stores and technical institutes. filecite turn3file4

For research publication, these go-to-market plans are best presented as deployment context rather than experimental evidence. Adoption targets should not be interpreted as achieved results unless actual analytics are available.

XIII. LIMITATIONS

- The current project material contains a strong product and prototype narrative but limited quantitative evidence for the effectiveness of the matching and AI components.
- The reported survey percentage requires a complete methodology before it can support generalizable research conclusions.
- The AI-agent concept requires explicit model implementation and evaluation if it is to be presented as an AI contribution.
- Market-size figures are based on project assumptions and should be externally sourced before publication.
- Identity and credential verification require careful treatment of privacy, security, consent, and applicable regulations.
- Real-world performance will depend on the quality, freshness, and completeness of professional profiles, availability information, service records, and customer feedback.

XIV. FUTURE WORK

The next phase of MISTRO can move the project from a prototype-centered system toward a quantitatively evaluated intelligent marketplace.

A. Learning-Based Ranking

Historical bookings can be used to learn professional-ranking patterns rather than relying only on manually selected weights.

B. Personalized Recommendations

The platform can learn customer preferences, repeat-service patterns, preferred time windows, and previous provider choices to improve recommendations.

C. Price Estimation

A data-driven price model can estimate expected service costs and flag unusually high or low quotes for further review.

D. Trust Modeling

A composite trust representation can incorporate verification status, review consistency, service history, cancellations, and other appropriate signals.

E. Predictive Service Allocation

The platform can eventually predict professional availability and likely service completion outcomes, helping reduce delays and unsuccessful bookings.

F. Broader Geographic Deployment

The project roadmap proposes expansion across Tamil Nadu, followed by additional states and service categories. □filecite□turn6file3□

XV. CONCLUSION

MISTRO presents a practical approach to a familiar but fragmented problem: finding a reliable person to handle a household service requirement. The proposed platform brings customers and skilled professionals into a common digital environment where service discovery, professional verification, matching, booking, communication, and reputation can be handled as connected steps rather than isolated activities.

The project places particular emphasis on trust and introduces a multi-factor matching concept based on skill, location, availability, and ratings. It also explores an AI-assisted layer covering service, price, trust, and verification. A prototype demonstrates the intended marketplace workflow, while preliminary survey material provides early evidence that the underlying customer problem is meaningful to the surveyed group.

The main research opportunity now lies in moving from a convincing product concept to measurable technical evidence. Implementing and evaluating the matching algorithm, defining reproducible AI components, collecting appropriate performance data, and conducting controlled usability experiments would allow MISTRO to be assessed not only as a marketplace idea but also as a technically evaluated system.

In that form, MISTRO can serve as a foundation for research into trust-aware local-service recommendation, professional reputation systems, and intelligent marketplace matching.

SOURCE AND PUBLICATION NOTE

This draft was prepared from the user-provided MISTRO project presentation titled “MISTRO — Service On Stress Gone / Sairam SDG Solveathon 4.0.” The presentation was treated as the primary source for project-specific facts, terminology, proposed business values, market assumptions, survey claims, prototype scope, and growth strategy. □filecite□turn3file3□

For an actual IEEE submission, the team should add independently verified scholarly and industry references for external claims, especially market statistics, competing-platform descriptions, trust/verification methods, recommendation systems, AI methods, and relevant sustainable-development literature. Any result reported as experimental should be backed by reproducible methodology and measured data.

Important: This is an original humanized manuscript draft, not a guarantee of zero similarity. Before submission, run the final paper through the institution or conference's approved similarity checker and verify every external citation.